

Lecture 22 — Kernel Modules and eBPF — Extending the Kernel

Jeff Zarnett
jzarnett@uwaterloo.ca

Department of Electrical and Computer Engineering
University of Waterloo

May 30, 2026

This lecture is not yet written. The outline lives in the plan file.

Intended sections:

- 1 Why extend the kernel?
- 2 Loadable Kernel Modules — anatomy
- 3 Why LKMs are dangerous
- 4 Module trust, signing, and the BYOVD problem
- 5 eBPF as the modern alternative
- 6 The eBPF verifier
- 7 Maps and the kernel/user boundary
- 8 Where eBPF is going
- 9 Monolithic vs. microkernel revisited